

A Statistically Supported DQN-Family Advantage in a Resource-Coupled Satellite-Ground Scheduling Benchmark

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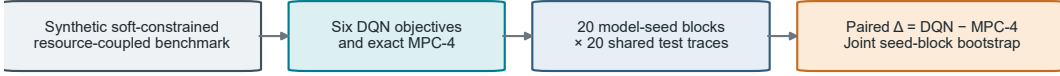
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Abstract

Resource-coupled satellite-ground scheduling requires decisions whose computational, energy, thermal, and communication consequences persist across task arrivals. We asked whether this temporal structure produces a reproducible advantage for the deep Q-network (DQN) family rather than for one selected objective. Six DQN objectives shared identical states, network architecture, replay, optimization, interaction budgets, 20 model seeds, and held-out traces across five fully coupled regimes of a fully specified synthetic, soft-constrained benchmark. Across all 30 objective–regime comparisons, every DQN mean was lower than MPC-4, every paired 95% bootstrap interval excluded zero, and all Holm-adjusted sign tests supported the ordering. The least-favorable mean difference was -5.92, and the family-wide simultaneous 95% upper bound was -5.03. Within-family differences were smaller and scenario dependent, whereas increasing exact planning depth from one to six steps reduced cost at substantially higher decision time. Together, these results establish a DQN-family advantage that is statistically supported across objective designs and resource-stress regimes, while providing a reproducible platform for future operational validation.

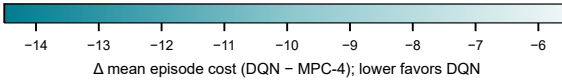
Keywords: deep Q-network variants; satellite edge computing; task scheduling; model-assisted reinforcement learning; full-action supervision.

a



b Mean episode-cost difference for every DQN objective and regime

Nominal	-12.5	-12.7	-13.0	-13.2	-12.4	-13.1
Burst	-11.0	-11.1	-11.1	-11.5	-10.7	-11.4
Link-limited	-13.5	-13.6	-13.8	-13.9	-13.1	-13.8
Energy-limited	-6.3	-6.2	-6.8	-7.0	-5.9	-6.9
Thermal-stress	-11.6	-11.8	-11.8	-12.1	-11.4	-12.1
	Standard DQN	Full-action Q auxiliary	Immediate advantage	Centered full-action	Double DQN	Double + centered



c Worst member remains below MPC-4

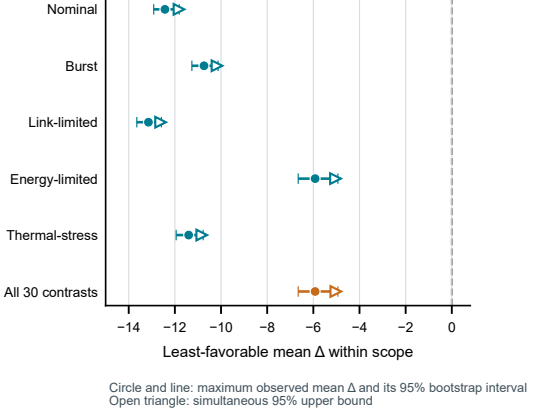


Figure 1: Family-level DQN–MPC-4 evidence in the synthetic soft-constrained benchmark. (a) Six DQN objectives and MPC-4 are evaluated in 20 paired model-seed blocks, each averaging 20 shared held-out traces. (b) Cells report the mean episode-cost difference, DQN minus MPC-4, for all 30 objective–regime contrasts; negative values favor DQN. (c) Circles and lines show the largest, least-favorable mean difference within each scope and its 95% joint-bootstrap interval; open triangles show simultaneous 95% upper bounds. A single 10,000-resample seed-block bootstrap preserves correlation among the 30 contrasts. Exact two-sided sign tests are Holm-adjusted across all 30 comparisons, all of which satisfy the prespecified support rule.

1 Introduction

Earth-observation and communication satellites increasingly execute perception and decision workloads in orbit. Compact networks can reduce downlink demand and response latency under constrained onboard resources [1–5]. However, onboard execution consumes heat, energy, and compute capacity, whereas ground offloading consumes contact time and bandwidth. Effective scheduling must therefore coordinate onboard execution, complete offloading, and collaborative preprocessing over time.

Existing schedulers optimize constellation coordination, observation plans, service placement, deadlines, makespan, or energy use [6–12]. Here, every decision changes six resources that persist across task arrivals: heat, energy, queue occupancy, compute utilization, bandwidth, and remaining contact. An inexpensive action can therefore restrict the choices available for later tasks. This temporal coupling motivates a central question: can learned long-horizon values outperform transparent one-step and finite-horizon alternatives across changing resource conditions?

Standard DQN [13,14] learns one Bellman target from each factual tuple (s_t, a_t, r_t, s_{t+1}) . In this setting, analytical resource equations can also evaluate every unexecuted action without advancing the environment or revealing later tasks. This structure enables a controlled test of both the DQN

family and alternative ways to use full-action supervision during training.

We compare six DQN objectives: standard DQN, full-action Q, immediate advantage, centered full-action, Double DQN, and Double DQN with centered full-action supervision. A separate $\gamma = 0$ contextual bandit provides a learned one-step comparator. Model-evaluated alternatives are available only during training; every deployed policy acts directly from its learned Q-values.

This study makes four contributions:

1. We introduce a fully specified, reproducible benchmark that links task primitives to three processing pathways and six persistent computational, energy, thermal, and communication resources.
2. We construct a controlled six-objective DQN family that varies factual supervision, full-action targets, centering, bootstrapping, and Double-DQN evaluation under matched training conditions.
3. We establish the family-level result directly against MPC-4 through 30 paired objective–regime contrasts, a correlation-preserving joint bootstrap, multiplicity-corrected tests, and a simultaneous upper bound.
4. We provide an executable evidence chain connecting simulator equations, transition audits, trained checkpoints, seed-level inference, generated tables, validation hashes, and publication figures.

2 Related Work and Study Positioning

2.1 Satellite-ground scheduling

Satellite schedulers cover constellation coordination, observation planning, service placement, and energy-aware offloading [6–12]. Our setting combines an explicit cost for every action with resource carryover between task arrivals. The one-step objective is therefore observed rather than hidden. The benchmark asks whether a learned value function can exploit consequences that persist beyond this observed one-step cost.

2.2 Model use and full-action learning

Dyna established predicted experience for value learning and planning [15], and fitted dynamics can accelerate deep Q-learning [16]. Counterfactual data fusion and causal augmentation construct transitions under structural assumptions [17, 18]. Counterfactual credit assignment instead separates action influence from future randomness [19]. Recent work also studies multi-action generative interfaces and joint quantities across action outcomes [20]. Together, these studies establish several routes for incorporating modeled alternatives into reinforcement learning.

Advantage learning widens action gaps through modified operators [21]. The centered full-action DQN objective instead retains the selected Bellman operator and adds supervised within-state action gaps. Table 1 distinguishes this construction from related mechanisms. Joint MDPs provide the closest multi-action interface [20], while full-action Q provides the closest empirical control. Our contribution is a controlled family-level evaluation that separates shared DQN performance from the smaller effects of target construction, centering, and bootstrapping.

In this manuscript, “counterfactual” denotes a model-evaluated one-step outcome for an unexecuted action under the same current task and exogenous state. It does not denote an identified causal effect from observational data.

Table 1: Feature comparison with the closest method families. “All actions” means that one learning update can use modeled outcomes for every discrete action at the sampled state.

Method family	Source of modeled information	How it enters learning	Relation to this study
Dyna and local model-based acceleration [15, 16]	Learned or fitted dynamics	Simulated experience or local rollouts	Does not use one simultaneous analytical label per action in the reported auxiliary loss
Causal counterfactual augmentation [18]	Structural assumptions and offline trajectories	Synthetic transitions augment the training distribution	Targets distributional coverage rather than within-state Bellman action gaps
Counterfactual credit assignment [19]	Future-conditioned trajectory information	Policy-gradient baseline or critic	Model-free and trajectory-conditioned rather than analytical one-step action enumeration
Joint MDPs [20]	Multi-action sample transition model	Joint action-outcome quantities and Bellman operators	Closest formal interface; it does not specify the centered DQN auxiliary loss evaluated here
This study	Deterministic analytical one-step transition for three actions	Auxiliary full-action Bellman targets; no replay insertion	Same real interactions as DQN, but three modeled action outcomes per eligible training state

3 System Model and Immediate-Cost Quantification

At task arrival t , a satellite selects one of three actions. Action α_s executes onboard and downlinks the result, whereas α_g downlinks the complete input for ground execution. Action $\alpha_{s\&g}$ performs onboard preprocessing followed by ground execution:

$$\mathcal{A} = \{\alpha_s, \alpha_g, \alpha_{s\&g}\}, \quad (1)$$

where the subscripts denote satellite, ground, and collaborative processing, respectively.

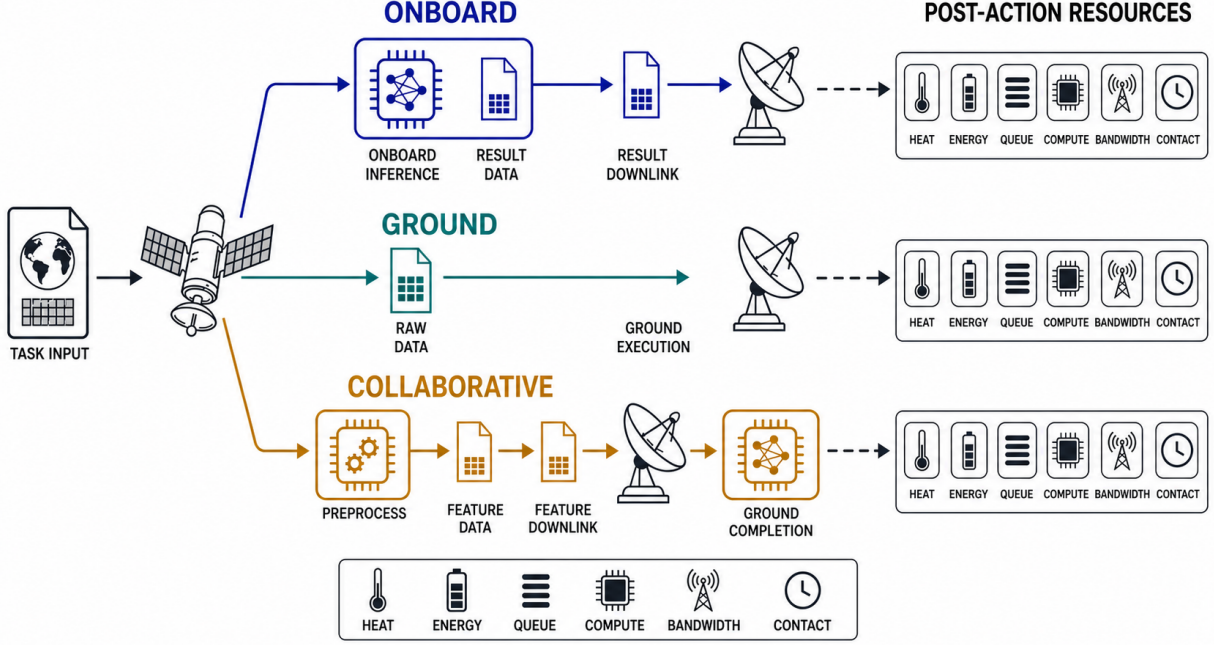


Figure 2: Three satellite-ground processing pathways. Each arriving task is executed onboard, transferred for ground execution, or preprocessed collaboratively before ground completion. The analytical transition model returns the immediate cost and the post-action heat, energy, queue occupancy, compute utilization, bandwidth, and remaining contact.

3.1 Implemented immediate-cost model

The experiments use the dimensionless linear cost implemented by the internally consistent synthetic simulator. Let x_k denote the 15 task descriptors in Table 2. Each descriptor is normalized as

$$\bar{x}_k = \text{clip}(x_k/s_k, 0, 1.5), \quad (2)$$

where s_k is the corresponding scale. For inherited normalized bandwidth b , the available return rate is $R(b) = 2.2 + 217.8b \text{ Mbit s}^{-1}$. The three action latencies are

$$\begin{aligned} L_s &= x_3 + 1000x_6/R(b), \\ L_g &= x_8 + 1000x_9/R(b), \\ L_{s\&g} &= 1000x_{12}/4 + x_{13} + 1000x_{15}/R(b). \end{aligned} \quad (3)$$

The implemented immediate costs are

$$\begin{aligned} c_s^{imm} &= 0.70\bar{x}_1 + 0.65\bar{x}_2 + 0.00045L_s + 0.55\bar{x}_5 + 0.35\bar{x}_6 + 0.35h + 0.12(1 - e), \\ c_g^{imm} &= 0.45\bar{x}_7 + 0.00045L_g + 0.55\bar{x}_9 + 0.35(0.28 - \tau)_+, \\ c_{s\&g}^{imm} &= 0.55(\bar{x}_{10} + \bar{x}_{11}) + 0.55\bar{x}_{14} + 0.00045L_{s\&g} + 0.45\bar{x}_{15} \\ &\quad + 0.18h + 0.12q + 0.18(0.22 - \tau)_+. \end{aligned} \quad (4)$$

Here h , e , q , and τ are inherited heat, remaining energy, queue occupancy, and contact duration. The weights are fixed simulator assumptions rather than fitted mission utilities. They are applied

identically during training, counterfactual preview, MPC evaluation, and deployment evaluation. Automated tests reproduce Eq. (4) from the 15 raw descriptors and compare it with all three code outputs.

Table 2: Implemented 15-dimensional task descriptor dictionary. The complete 24-dimensional observation appends normalized time, sine and cosine of link phase, and six resource coordinates: h , e , q , compute utilization u , bandwidth b , and contact duration τ .

Coordinate	Descriptor	Unit	Normalization scale s_k
x_1	Onboard compute heat	J	100
x_2	Total workload	GFLOP	4.8
x_3	Onboard compute time	ms	1400
x_4	Result transmit time at 50 Mbit s ⁻¹	ms	100
x_5	Squared co-location count	count ²	2401
x_6	Onboard-result data	Mbit	4
x_7	Full-offload transmit heat	J	100
x_8	Ground compute time	ms	100
x_9	Raw input data	Mbit	76.8
x_{10}	Preprocessing heat	J	100
x_{11}	Feature-transmit heat	J	100
x_{12}	Onboard preprocessing work	GFLOP	2.2
x_{13}	Remaining ground compute time	ms	100
x_{14}	Total workload descriptor	GFLOP	4.8
x_{15}	Collaborative feature data	Mbit	32

3.2 Internally consistent task and communication generator

The simulator does not sample heat, workload, data volume, and latency independently. It first draws correlated latent variables (v_t, w_t) with correlation parameter 0.68 and maps them through a logistic transform to raw data volume $D_t \in [8, 64]$ Mbit and workload $W_t \in [0.25, 4.0]$ GFLOP. Result and collaborative-feature volumes are

$$D_t^{res} = r_t D_t, \quad r_t \in [0.01, 0.05], \quad D_t^{feat} = f_t D_t, \quad f_t \in [0.08, 0.30]. \quad (5)$$

The implementation enforces $D_t^{res} \leq D_t^{feat} \leq D_t$. A workload-dependent fraction $p_t \in [0.20, 0.45]$ is processed onboard in the collaborative mode. Given onboard and ground rates $R_s = 4$ and $R_g = 80$ GFLOP s⁻¹, processing time is derived as W/R , rather than sampled separately. Compute power rises monotonically from 4 to 20 W with workload; compute heat is 0.78 times compute energy. Transmission time is data volume divided by the effective return rate

$$R_t^{link} = 2.2 + 217.8b_t \quad \text{Mbit s}^{-1}, \quad (6)$$

and transmission heat is 0.55 times transmit energy. These values are engineering envelopes, not fitted flight distributions. Their scale is consistent with modern SmallSat computing-system power reported by NASA. NASA ground-network examples span 2.2 Mbit s⁻¹ S-band to 220 Mbit s⁻¹ X-band returns [22, 23]. The bounded data-reduction factors represent the bandwidth-reduction

role of onboard image compression, not a claim about one instrument-specific codec [24]. Table 3 states every implemented assumption.

Table 3: Modeled engineering primitives and provenance of the current simulator. Values define a transparent synthetic envelope; they are not presented as a fitted flight distribution.

Quantity	Implemented range	Basis
Raw task data	8–64 Mbit nominal	Stress interval reaches 92.8 Mbit before sensitivity scaling
Arithmetic workload	0.25–4.0 GFLOP nominal	Stress interval reaches 5.8 GFLOP
Result / raw-data ratio	1–5%	Bounded compression assumption
Feature / raw-data ratio	8–30%	Bounded collaborative-processing assumption
Preprocessing fraction	20–45%	Deterministic function of workload quantile
Onboard / ground rate	4 / 80 GFLOP s ⁻¹	Fixed simulator hardware envelope
Compute power	4–20 W	Bracketed by NASA SmallSat avionics examples
Effective RF return rate	2.2–220 Mbit s ⁻¹	NASA SmallSat ground-system examples
Transmit power	6 W	Fixed simulator assumption
Burst multiplier	1.45×	Contiguous prespecified stressor

An automated audit generated 20,000 tasks before training. All values were finite and positive, all data-flow inequalities held, and the empirical raw-data–workload correlation was 0.660. Figure 3 shows the resulting dependence structure. Constant-rate columns have zero variance and are therefore excluded from correlation coefficients.

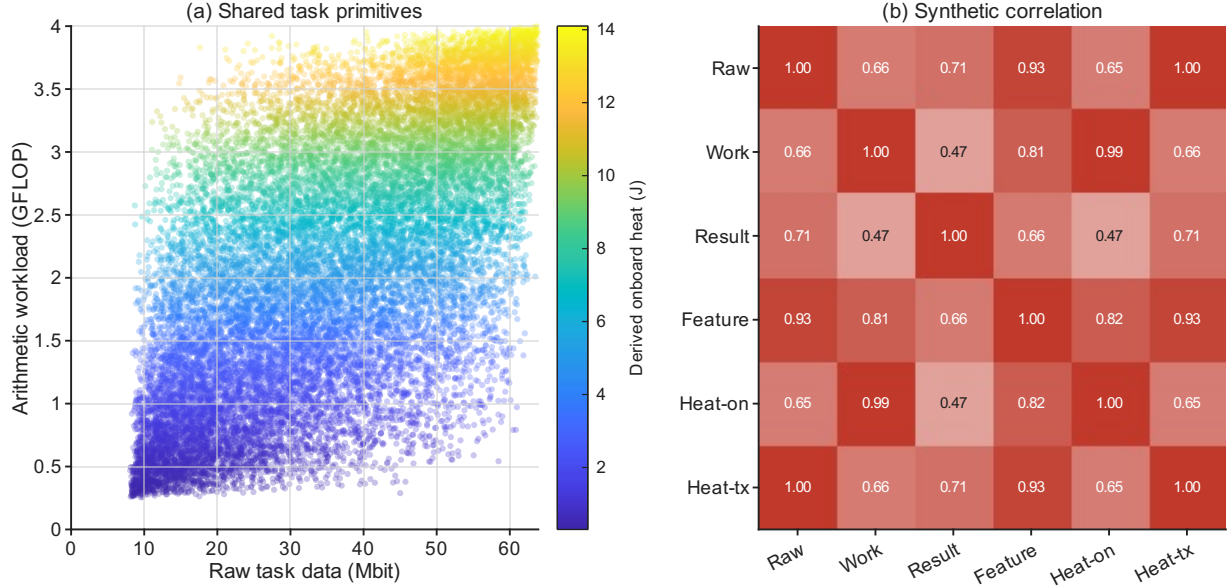


Figure 3: Internal-consistency audit of the synthetic generator. (a) A legibility sample of 2,000 of 20,000 tasks; color encodes heat derived from compute energy. (b) Correlations computed from all 20,000 tasks. No fitted flight distribution is implied.

3.3 Complete stochastic generator and action-load specification

For independent reconstruction, this subsection states the remaining simulator functions that enter the transition. Let $u_D, \epsilon_W \sim \mathcal{N}(0, 1)$ independently, $u_W = 0.68u_D + \sqrt{1 - 0.68^2}\epsilon_W$, and $g(u) =$

$[1 + \exp(-1.55u)]^{-1}$. Nominal raw volume and workload are

$$D = 8 + 56g(u_D) \text{ Mbit}, \quad W = 0.25 + 3.75g(u_W) \text{ GFLOP}. \quad (7)$$

The burst interval occupies indices $\lfloor T/3 \rfloor$ through $\lfloor T/3 \rfloor + \max(4, \lfloor T/4 \rfloor) - 1$ and multiplies both quantities by 1.45. With independent $U_r, U_f \sim \mathcal{U}(0, 1)$,

$$\begin{aligned} r &= 0.01 + 0.04U_r, & f &= 0.08 + 0.22 \text{clip}[0.65g(u_W) + 0.35U_f, 0, 1], \\ p &= 0.20 + 0.25g(u_W), & D^{res} &= rD, \quad D^{feat} = fD, \quad W^{pre} = pW, \\ n &= \text{round}\{6 + 43 \text{clip}[0.60g(u_W) + 0.40g(u_D), 0, 1]\}, & x_5 &= n^2. \end{aligned} \quad (8)$$

Onboard and ground rates are 4 and 80 GFLOP s^{-1} , respectively. Compute power is $P_c = 4 + 16 \text{clip}(W/4.8, 0, 1)^{0.70}$ W, and transmit power is 6 W. Compute and transmit heat are 0.78 and 0.55 times their corresponding energy use. These definitions generate all 15 coordinates in Table 2.

At reset, a single phase $\phi \sim \mathcal{U}(0, 2\pi)$ generates the exogenous link trace

$$\begin{aligned} \bar{b}_t &= \text{clip} \left\{ \kappa_b [0.62 + 0.23 \sin(2\pi t/24 + \phi) + \epsilon_t^b] - I_{link}, 0.12, 1 \right\}, \\ \bar{\tau}_t &= \text{clip} \left\{ 0.55 + 0.35 \sin(2\pi t/31 + \phi/2) + \epsilon_t^\tau, 0.08, 1 \right\}, \end{aligned} \quad (9)$$

where $\epsilon_t^b \sim \mathcal{N}(0, 0.06^2)$, $\epsilon_t^\tau \sim \mathcal{N}(0, 0.04^2)$, κ_b is the sensitivity multiplier, and $I_{link} = 0.22$ only in the link-limited regime. All episode tasks and link states are drawn at reset from one seeded NumPy generator.

The normalized action loads used in Eq. (12) are stated here in action order $(\alpha_s, \alpha_g, \alpha_{s\&g})$:

$$\begin{aligned} \hat{w} &= (W, 0.05W, W)/4.8, & \hat{Q} &= (x_1, 0.20x_7, x_{10} + x_{11})/100, \\ \hat{B} &= (x_6, x_9, x_{15})/76.8, \\ \Delta e &= \kappa_e \left(0.012 + 0.042\hat{w}_1, 0.007 + 0.025\hat{B}_2/b, 0.010 + 0.025\hat{w}_3 + 0.018\hat{B}_3/b \right), \\ \Delta q &= \left(0.055\hat{w}_1(1 + u), 0.048\hat{B}_2/b, 0.030(\hat{w}_3 + \hat{B}_3/b) \right), \\ \mu &= (0.030(1 - u), 0.035b\tau, 0.032(1 - 0.5u)), \\ \ell &= (0.02, 0.18\hat{B}_2, 0.12\hat{B}_3). \end{aligned} \quad (10)$$

Here divisions by b use $\max(b, 0.1)$. The utilization targets in Eq. (12) are $(\hat{w}_1, 0.10\hat{B}_2, 0.55\hat{w}_3)$. Initial resources are $(h, e, q, u) = (0.22, 0.88, 0.12, 0.18)$ except that energy-limited episodes use $e = 0.62$ and thermal-stress episodes use $h = 0.42$. Each episode terminates after exactly 64 executed actions; the terminal Bellman multiplier is zero. Energy and other resources remain soft-constrained clipped accounting states. In particular, $e = 0$ neither masks infeasible actions nor terminates an episode. It therefore does not represent a conserved battery state, and clipping does not retain the magnitude of an energy deficit.

4 Sequential Scheduling Formulation

4.1 State and transition

The normalized 24-dimensional state is

$$s_t = [x_t, \eta_t, z_t], \quad \eta_t = [t/(T - 1), \sin \phi, \cos \phi], \quad z_t = [h_t, e_t, q_t, u_t, b_t, \tau_t], \quad (11)$$

where $x_t \in \mathbb{R}^{15}$ contains the task inputs defined in Table 2. The context vector η_t exposes absolute episode time and the reset-level link phase that determine the time-varying task and link distributions. The remaining task and link innovations are independent draws conditional on this context. The six resource variables denote normalized heat, remaining energy, queue occupancy, compute utilization, bandwidth, and contact duration. Thus, within each fixed regime, the transition distribution is conditioned on all persistent simulator variables rather than on a hidden reset phase.

Executing action a_t changes resources before the next exogenous task arrives. The implemented transition is

$$\begin{aligned} h_{t+1} &= \text{clip}(0.86h_t + 0.17c\widehat{Q}_t^{(a_t)}, 0, 1.3), \\ e_{t+1} &= \text{clip}(e_t + 0.006 - c\Delta e_t^{(a_t)}, 0, 1), \\ q_{t+1} &= \text{clip}(q_t + c(\Delta q_t^{(a_t)} - \mu_t^{(a_t)}), 0, 1.3), \\ u_{t+1} &= \text{clip}(0.68u_t + 0.32c\widehat{w}_t^{(a_t)}, 0, 1.2), \\ b_{t+1} &= \text{clip}(\bar{b}_{t+1} - c\ell_t^{(a_t)}, 0.08, 1), \\ \tau_{t+1} &= \text{clip}(\bar{\tau}_{t+1} - 0.10c\widehat{B}_t^{(a_t)}, 0.05, 1). \end{aligned} \tag{12}$$

Here $\widehat{Q}$, $\widehat{w}$, and $\widehat{B}$ are normalized action-specific heat, work, and data demands. The terms Δe , Δq , μ , and ℓ are the deterministic resource-use, queue, service, and link-occupation functions defined in Eq. (10). Exogenous link variables $\bar{b}$ and $\bar{\tau}$ follow Eq. (9) and are sampled before each episode. Coupling $c = 0$ removes action-dependent carryover, $c = 0.5$ weakens it, and $c = 1$ defines full coupling.

4.2 Reward and standard DQN backbone

The reward is the negative immediate cost plus post-decision feasibility penalties:

$$\begin{aligned} r_t = & -c_t^{imm} - c[12(h_{t+1} - 0.78)_+ + 15(0.18 - e_{t+1})_+ \\ & + 8(q_{t+1} - 0.75)_+ + 10\mathbf{1}_{a_t \neq \alpha_s}(0.14 - \tau_t)_+]. \end{aligned} \tag{13}$$

The DQN uses the unmodified factual target

$$y_t = r_t + \gamma(1 - d_t) \max_{a'} Q_{\bar{\theta}}(s_{t+1}, a'), \tag{14}$$

and factual loss

$$L_{DQN} = \mathcal{H}(Q_{\theta}(s_t, a_t), y_t), \tag{15}$$

where $\mathcal{H}$ denotes the Huber loss. The target network both selects and evaluates the maximizing action, as required by standard DQN.

5 DQN Variants with Model-Evaluated Full-Action Targets

The DQN family shares the factual target and loss in Eq. (14). Its model-assisted variants add a training-only auxiliary path that can supply labels for unexecuted actions. The following construction defines the centered full-action member; the experimental design then removes centering, removes the bootstrap, or changes target evaluation to measure how these choices alter performance.

5.1 Side-effect-free full-action preview

For every replayed factual state, the analytical transition evaluates each $a \in \mathcal{A}$:

$$(\tilde{\ell}_{t,a}, \tilde{z}'_{t,a}) = F_{\text{preview}}(x_t, z_t, a). \quad (16)$$

Here $\tilde{\ell}_{t,a}$ contains the immediate cost and post-decision feasibility penalty for action a . The preview does not advance time, mutate state, alter trajectories, or consume random numbers. Before action selection it computes only the current-task cost and six post-action resources. The factual action is then executed once, after which the next exogenous task x_{t+1} is revealed and the replay tuple is completed. During a later replay update, all alternatives reuse this observed next task and exogenous link trace. The counterfactual next state therefore replaces only the six endogenous resource coordinates:

$$\tilde{s}'_{t,a} = [x_{t+1}, \eta_{t+1}, \tilde{z}'_{t,a}]. \quad (17)$$

Evaluation and deployment never call the preview interface.

5.2 All-action counterfactual Bellman targets

The simulator-exact one-step outcome yields

$$\tilde{y}_{t,a} = -\tilde{\ell}_{t,a} + \gamma(1 - d_t) \max_{a'} Q_{\bar{\theta}}(\tilde{s}'_{t,a}, a'). \quad (18)$$

The targets create no multi-step rollout and insert no counterfactual transition into replay. They are used only after state-wise centering in the complete method below.

5.3 Centered advantage distillation

Absolute targets in Eq. (18) share a bootstrapped state-value offset. Deployment selects $\arg \max_a Q(s, a)$, so a common additive offset cannot change the action. We define predicted and target centered advantages as

$$\begin{aligned} \hat{A}_{t,a} &= Q_{\theta}(s_t, a) - \frac{1}{|\mathcal{A}|} \sum_j Q_{\theta}(s_t, j), \\ \tilde{A}_{t,a} &= \tilde{y}_{t,a} - \frac{1}{|\mathcal{A}|} \sum_j \tilde{y}_{t,j}. \end{aligned} \quad (19)$$

The advantage and total losses are

$$\begin{aligned} L_{adv} &= \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \mathcal{H}(\hat{A}_{t,a}, \tilde{A}_{t,a}), \\ L_{CFA} &= L_{DQN} + \lambda L_{adv}. \end{aligned} \quad (20)$$

The factual term anchors the absolute Q scale, while L_{adv} teaches all action gaps. The only added algorithmic hyperparameter is λ .

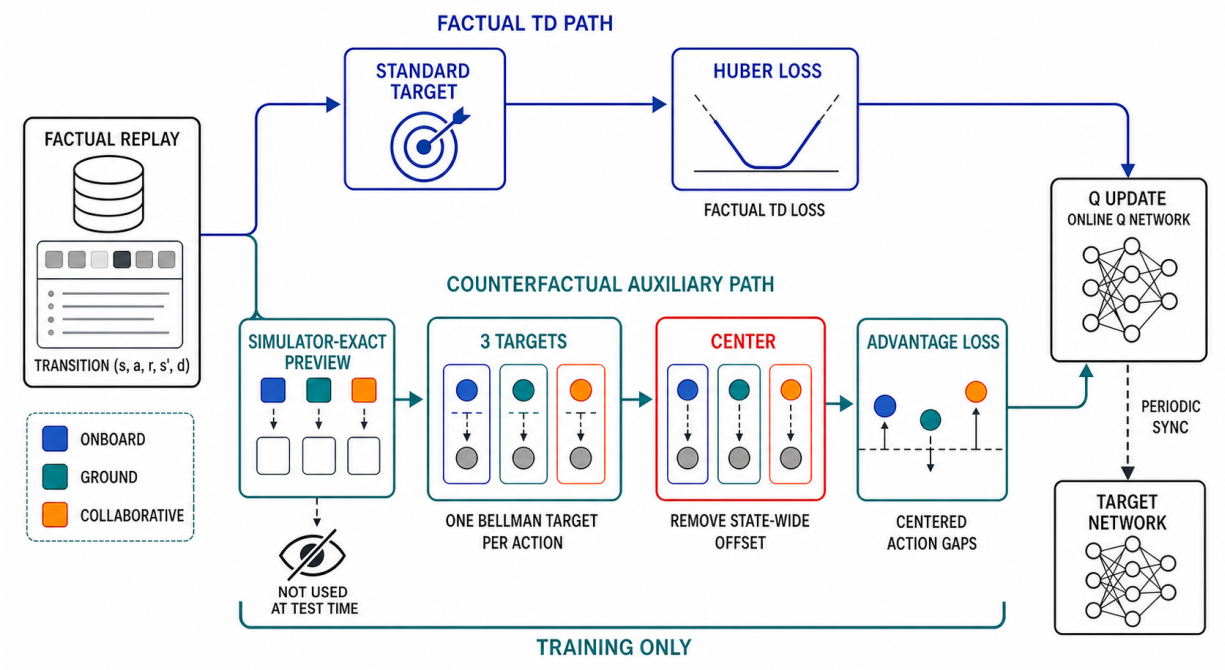


Figure 4: Training-only centered full-action mechanism. The upper path retains the factual standard-DQN target and Huber loss. The lower path previews all three actions, constructs three Bellman targets, centers them within state, and distills their relative advantages. Both losses update the online Q-network; the target network is synchronized only at the shared fixed interval. Preview outputs are unavailable during evaluation and deployment.

Algorithm 1: DQN training with centered full-action supervision

1. Initialize uniform replay $\mathcal{D}$, online Q_θ , and target $Q_{\bar{\theta}} \leftarrow Q_\theta$.
2. At each real step, observe s_t and preview current-task outcomes $(\tilde{\ell}_{t,a}, \tilde{z}'_{t,a})$ for all actions without side effects.
3. Select a_t by the shared epsilon-greedy schedule and execute it once. After x_{t+1} is revealed, store the factual tuple plus the previously computed previews.
4. When an update is due, sample one uniform mini-batch, compute the factual loss, compute Eqs. (18) and (19), and update θ using Eq. (20).
5. Synchronize $Q_{\bar{\theta}}$ at the shared fixed interval.

Training previews are never used for evaluation or deployment.

6 Experimental Design and Settings

6.1 Design principles

The design addresses four comparisons. Standard DQN versus its centered full-action variant measures the effect of adding the complete auxiliary objective to an unchanged factual DQN backbone.

Uncentered full-action Q and centered immediate-cost targets receive the same three modeled outcomes per state. They match the type and count of model outputs. They do not by themselves isolate centering or Bellman bootstrapping because their target scales differ and their auxiliary weights were not tuned separately. Double DQN with and without centered full-action supervision assesses whether the fixed auxiliary implementation has a similar descriptive effect when action selection and target evaluation are decoupled [25]. The $c \in \{0, 0.5, 1\}$ regimes and planners with $H \in \{1, 2, 4, 6\}$ bound the family comparison as the implemented objective and planning depth change.

Standard DQN and its centered full-action variant share every component except the auxiliary objective. They use the same input, network, replay, exploration, optimizer, update cadence, target synchronization, and gradient clipping. The full-action Q control applies the same auxiliary weight directly to the three uncentered Bellman targets. The immediate-advantage control centers the negative analytical one-step costs but omits their bootstrapped future values. Both Double-DQN variants use online-network action selection with target-network evaluation in factual and auxiliary Bellman targets. Dueling networks, prioritized replay, multi-step returns, noisy networks, and distributional value learning remain outside the comparison.

The auxiliary weight was fixed before the internally consistent simulator experiment. A preceding calibration with model seeds 700–703 compared $\lambda \in \{0.1, 0.3, 1, 3\}$. It selected $\lambda = 0.3$ by maximizing the worst relative improvement of the centered full-action variant across five coupled regimes, with the smaller weight as the tie-break. This selection rule was specific to the centered objective. The uncentered full-action, immediate-advantage, and Double-DQN auxiliary variants inherited $\lambda = 0.3$ without method-specific calibration. Because auxiliary losses differ in scale and error structure, their comparisons are fixed-coefficient implementation comparisons rather than a factorial identification of centering or bootstrapping. The calibration used an earlier simulator version and is a prior specification rather than evidence from the current generator. Confirmation and all component comparisons use seeds 800–819 without post hoc removal, replacement, or selection. Every learned policy was trained only on the nominal $c = 1$ distribution.

6.2 Training configuration

Each network has 24 inputs, hidden widths 128–128–64, ReLU activations, and three outputs. Every model receives 600 episodes of 64 tasks, totaling 38,400 real interactions. Fixed settings are $\gamma = 0.97$, learning rate 3×10^{-4} , batch size 128, replay capacity 100,000, and gradient-norm limit 10. Training begins after 128 transitions and performs one update every four interactions. The checkpoint field `target_interval=250` is evaluated only on update calls; because updates occur at interaction counts divisible by four, synchronization occurs every 500 real interactions, equivalent to 125 optimizer updates. Epsilon decreases linearly from 1 to 0.05 over 13,440 interactions.

The contextual bandit uses the same network, optimizer, exploration schedule, and interaction budget, with $\gamma = 0$. It therefore learns the modeled one-step reward without a long-horizon Bellman term. Uniform replay stores only one-step factual transitions, with preview outputs attached as auxiliary training data. Standard DQN and Double DQN use no analytical alternative-action labels. The four auxiliary variants evaluate three modeled outcomes at every real interaction. The checkpoint manifest records these counts, optimizer updates, and wall time for every trained model.

All reported experiments use the internally consistent synthetic generator described above. Link bandwidth and contact follow bounded noisy periodic traces generated at reset. Burst and thermal-stress episodes multiply raw data and workload by 1.45 during one contiguous interval; all dependent heat, energy, latency, and transmission quantities are recomputed from those primitives. An earlier independently sampled generator is retained only as a historical reproducibility mode and is not

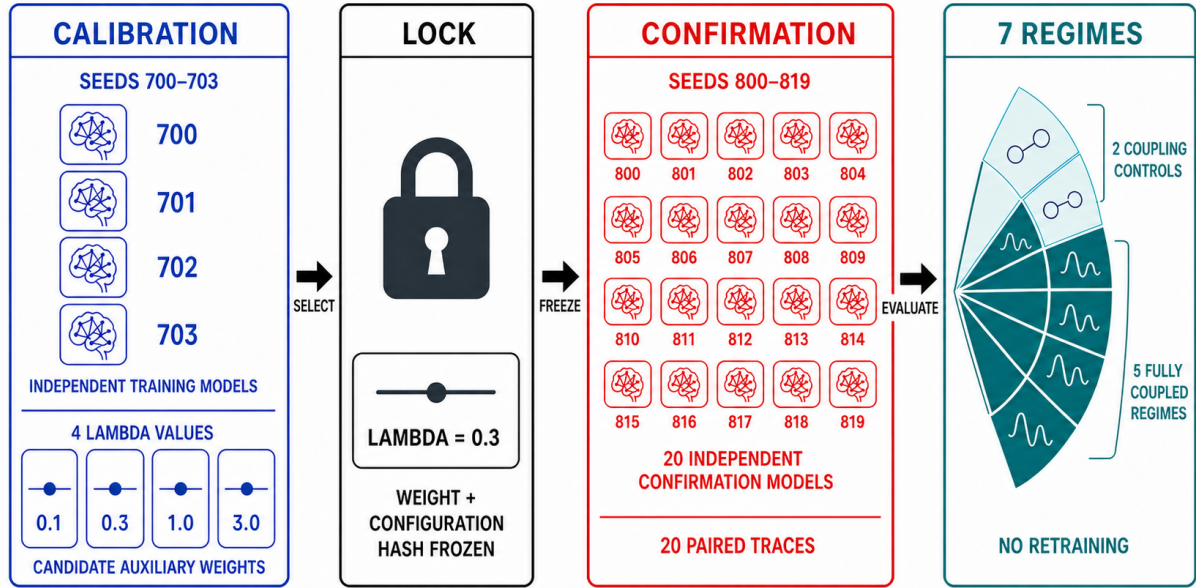


Figure 5: Prespecification and confirmation protocol. An earlier-simulator calibration with four seeds selected $\lambda = 0.3$ for the centered full-action objective before the current synthetic experiment. Other auxiliary objectives inherited this coefficient without method-specific tuning. Twenty disjoint confirmation seeds, 800–819, are each evaluated on 20 shared held-out traces in seven regimes, with no current-generator tuning or seed exclusion.

used for the reported results.

6.3 Seven evaluation regimes

The seven regimes include two composite coupling-boundary conditions and five fully coupled operating conditions:

1. **Static** ($c = 0$): removes action-dependent state transitions and scales delayed feasibility penalties to zero.
2. **Reduced coupling** ($c = 0.5$): halves action-dependent resource effects and the delayed feasibility-penalty scale.
3. **Nominal** ($c = 1$): matches the fully coupled training distribution.
4. **Burst**: increases contiguous compute and thermal demand.
5. **Link-limited**: reduces the available bandwidth trace by 0.22 before clipping.
6. **Energy-limited**: lowers initial energy from 0.88 to 0.62.
7. **Thermal-stress**: raises initial heat from 0.22 to 0.42 and includes the burst interval.

The first two regimes are composite interventions because the same c appears in both the transition in Eq. (12) and the delayed-penalty term in Eq. (13). They therefore do not isolate action-dependent carryover from reward rescaling. The remaining five regimes retain $c = 1$. Models trained at $c = 1$ are transferred to the two boundary conditions without retraining. These rows measure behavior under a joint objective-and-dynamics shift rather than the best policy attainable at lower coupling.

6.4 Planning comparator, outcomes, and inference

The deployment comparison includes immediate-cost argmin, a contextual bandit, a prespecified resource-threshold heuristic, standard DQN, centered full-action DQN, and an exact H -step receding-horizon model-based planner. The threshold rule selects ground processing when $h > 0.72$ or $e < 0.24$ and both $b > 0.28$ and $\tau > 0.18$; under the same heat or energy condition it otherwise selects collaborative processing. If that condition is absent, it selects onboard processing when $b < 0.22$ or $\tau < 0.14$, collaborative processing when $q > 0.70$, and immediate-cost argmin otherwise. These thresholds were hand specified and not fitted to the evaluation traces. Fixed onboard and fixed ground policies always select α_s and α_g , respectively. At every real decision, the planner enumerates all 3^H action sequences over the next H known tasks and link states. It minimizes discounted modeled cost with $\gamma = 0.97$, executes only the first action, and replans. The primary table retains $H = 4$. A separate nominal analysis evaluates $H \in \{1, 2, 4, 6\}$ on 400 unique held-out traces and reports wall-clock cost per decision. Each planner is exact only for its stated perfect-forecast objective, not for the 64-step global problem.

The primary outcome is the undiscounted episode total cost

$$C_{\text{episode}} = \sum_{t=0}^{T-1} (c_t^{\text{imm}} + p_t) = - \sum_{t=0}^{T-1} r_t, \quad (21)$$

where p_t is the complete nonnegative penalty inside the square brackets in Eq. (13), including its multiplier c . Every executed step, including the terminal step, contributes once. DQN and

the finite-horizon planner optimize discounted targets with $\gamma = 0.97$, whereas Eq. (21) reports accumulated operational cost without discounting. This optimization–evaluation distinction was not subjected to a separate discount sensitivity analysis. Secondary outcomes are immediate cost, delayed penalty, energy use, modeled latency, transmitted data, threshold-exposure steps, planner wall-clock time, maximum heat, and minimum remaining energy.

The independent inferential unit is the model-seed block. For learned–learned comparisons, the same seed initializes network weights and controls episode task seeds, exploration draws, and replay-index sampling. Trajectories may diverge after different greedy actions, but the matched seed remains the randomization block. Each seed is evaluated on the same 20 held-out traces, which are averaged within seed, giving $n = 20$ blocks per learned method. For the central DQN-family–MPC-4 claim, every raw row is matched on regime, model-seed trace namespace, and trace seed before trace-level differences are averaged within seed. MPC-4 is deterministic conditional on a trace; the model-seed field therefore indexes its matched trace namespace rather than a trained planner replicate.

All paired bootstrap analyses use 10,000 resamples with seed 20260829. Individual two-sided 95% intervals are percentile intervals, and exact two-sided paired sign tests assess directional consistency. For the 30 DQN-objective–regime contrasts against MPC-4, one joint bootstrap resamples the same 20 seed blocks across all contrasts, preserving their correlation. Sign-test values are Holm-adjusted across all 30 cells. We additionally report the maximum, least-favorable mean difference within each regime and across all 30 cells. A simultaneous one-sided 95% upper bound adds a joint critical value to every observed cell mean. The critical value is the 95th percentile of the largest centered bootstrap deviation across the relevant cells. An upper bound below zero supports the joint statement that every mean difference is negative. For within-family contrasts, Holm adjustment is applied across the five fully coupled regimes separately for each prespecified contrast. The eight parameter profiles form a separate family, and each preview-scale contrast forms its own five-regime family. A directional cell is labelled supported only when its mean favors the first-named method, its paired interval excludes zero, and its family-adjusted sign-test value is below 0.05. We report seed-level differences and leave-one-seed-out ranges for the primary centered-full-action contrast. Its conservative sensitivity analysis resamples the two sets of 20 seed means independently and ignores matching. Test traces and bootstrap draws are never treated as independent model replicates.

6.5 Prespecified sensitivity and training diagnostics

Without retraining or model selection, we repeated the nominal evaluation under eight profiles. They covered the reference configuration, compute demand $\pm 20\%$, raw data volume $\pm 20\%$, energy use $+20\%$, heat load $+20\%$, and link capacity -20% . Each profile retained the same 20 models and the same held-out trace namespace as the primary experiment, with Holm adjustment across the eight profiles. Training logs record every episode for all 20 seeds. Curves report a causal 25-episode moving mean and seed-level mean $\pm$ SD for total cost, energy use, latency, and delayed penalty. These diagnostics describe optimization trajectories; confirmation claims remain based on held-out traces.

6.6 Preview-model misspecification test

The main experiment assumes that the analytical preview matches the simulator transition. To test this dependence, two additional centered full-action model sets were trained without changing the factual environment or deployment evaluation. One set scaled preview one-step costs (immediate

cost plus modeled penalty) and action-induced resource changes by 0.85, while the other used 1.15. Resource predictions were clipped to the same physical bounds after scaling around the current resource state. Each condition used the same 20 seeds, training budget, network, replay, and held-out trace namespace as the main experiment. These scales were fixed as symmetric stress tests and were not used for model selection.

6.7 Reproducible result entry point

The core result entry point is `python -m md_cbad_dqn.viewer_experiments all --seeds 800-819 --resume`. Component and Double-DQN comparisons use `python -m md_cbad_dqn.extended_revision --seeds 800-819 --stage all --resume`. The first command trains or loads 60 primary learned-policy checkpoints plus 40 preview-error checkpoints. The second reuses the primary DQN checkpoints and trains 80 component and Double-DQN checkpoints. It then evaluates six DQN objectives, performs the independent-seed bootstrap, and evaluates four planning depths on 400 unique traces. The command `python -m md_cbad_dqn.extended_reporting` recomputes the 30 paired DQN-MPC-4 contrasts, joint maximum statistics, LaTeX tables, and artifact hashes from locked raw CSV files. The command `python -m md_cbad_dqn.transition_audit` writes a fixed-seed, all-action example in which preview and executed costs and six resources agree coordinate by coordinate. Unit tests repeat this equality at both resource clipping boundaries and on the terminal step, and verify that time and link phase occupy the three added state coordinates. Experiments used Python 3.13.9, NumPy 2.4.1, pandas 3.0.3, and PyTorch 2.8.0 with CUDA 12.9. Hardware comprised an NVIDIA GeForce RTX 5070 Ti running Windows 11. Planning wall time was measured on a 12th Gen Intel Core i9-12900K with 24 logical processors after one untimed warm-up episode. Values report the mean over all timed decisions. Machine-readable metadata record the 24-dimensional state definition, resolved variants, seeds, budgets, randomization blocks, model-information counts, result files, and runtime environment. Table and figure scripts consume these files directly.

7 Results

7.1 Direct paired inference establishes a DQN-family advantage under full coupling

We first tested the family-level claim against the strongest implemented non-DQN comparator. MPC-4 had the lowest non-DQN mean in every fully coupled regime. All 30 paired objective-regime comparisons favored DQN, and every individual 95% bootstrap interval excluded zero.

Table 4: Episode total cost under the internally consistent synthetic generator. For learned policies, SD is across 20 independently trained model-seed blocks after within-seed trace averaging. For deterministic nonlearned policies, SD is across the 20 disjoint trace namespaces and does not represent model-training variability. These two SDs quantify different sources of variation and are not directly comparable as inferential uncertainty. All learned policies were trained only at nominal $c = 1$; static and reduced-coupling rows are transfers without retraining. MPC-4 is exact only for its stated four-step perfect-forecast objective. Lower is better.

Regime	Argmin	MPC-4	Bandit	Threshold	DQN	Centered FA
Static ($c = 0$)	24.55 ± 0.21	24.55 ± 0.21	24.78 ± 0.33	24.74 ± 0.25	26.85 ± 0.99	26.66 ± 0.64
Reduced ($c = 0.5$)	26.65 ± 0.63	25.70 ± 0.26	26.39 ± 1.05	26.43 ± 0.38	27.81 ± 0.68	27.61 ± 0.56
Nominal	193.42 ± 8.33	90.83 ± 1.71	101.00 ± 4.94	188.08 ± 6.41	78.33 ± 1.88	77.66 ± 1.84
Burst	241.20 ± 6.84	108.35 ± 1.99	136.55 ± 11.20	238.74 ± 6.56	97.36 ± 2.01	96.88 ± 1.94
Link-limited	272.02 ± 7.98	115.73 ± 1.92	139.75 ± 7.36	227.25 ± 8.36	102.26 ± 1.75	101.80 ± 1.80
Energy-limited	225.31 ± 7.73	126.00 ± 1.58	133.26 ± 4.20	246.46 ± 6.45	119.67 ± 2.14	119.00 ± 1.65
Thermal-stress	247.58 ± 6.25	109.20 ± 2.01	137.74 ± 10.79	241.80 ± 6.26	97.55 ± 2.03	97.10 ± 1.93

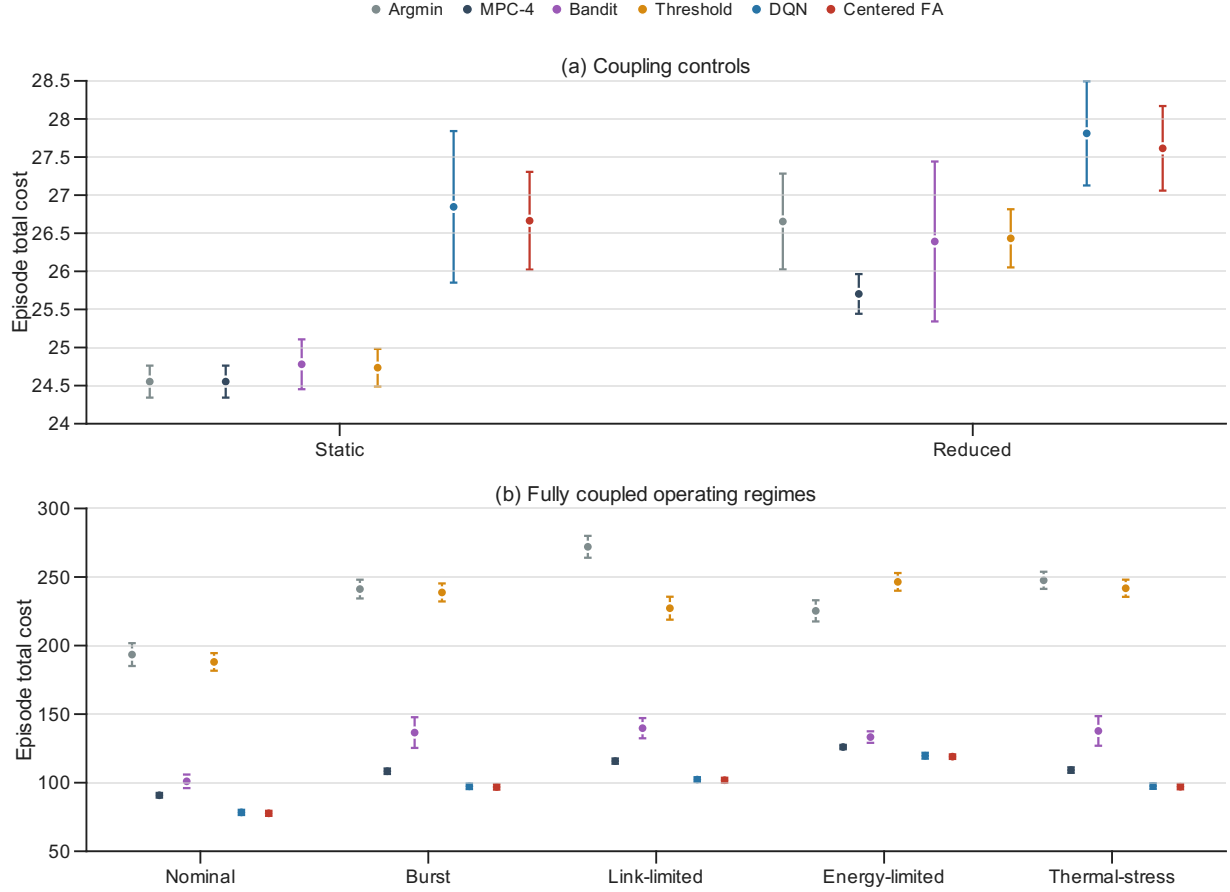


Figure 6: Seven-regime performance under internally consistent synthetic task generation. For learned policies, error bars are SD across 20 model-seed blocks after within-seed trace averaging. Deterministic comparators instead show variation across trace namespaces, so their error bars are descriptive and not inferentially interchangeable with those of learned methods. All learned models were trained at nominal $c = 1$; the $c < 1$ rows are transfers without retraining. The ordinate changes between panels and panel heights should not be compared directly.

Table 5: Direct joint inference for the DQN family relative to MPC-4. Differences are DQN minus MPC-4, so negative values favor DQN. Within each regime, the table reports the largest, least-favorable mean among six DQN-objective contrasts; the final row takes the maximum across all 30 objective-regime cells. Intervals are percentile intervals for the corresponding maximum from 10,000 joint bootstrap resamples. Simultaneous one-sided 95% upper bounds use the maximum centered bootstrap deviation. The independent unit is the model-seed block ($n = 20$), after averaging 20 shared held-out traces within each block.

Scope	Least-favorable DQN objective	Maximum Δ	95% interval for maximum	Simultaneous 95% upper bound
Nominal	Double DQN	-12.43	[-12.92, -11.81]	-11.82
Burst	Double DQN	-10.74	[-11.26, -10.13]	-10.17
Link-limited	Double DQN	-13.15	[-13.65, -12.58]	-12.62
Energy-limited	Double DQN	-5.92	[-6.66, -4.94]	-5.04
Thermal-stress	Double DQN	-11.40	[-11.94, -10.77]	-10.84
All five regimes	Double DQN (Energy-limited)	-5.92	[-6.66, -4.94]	-5.03

The joint seed-block analysis supported the same conclusion after controlling multiplicity across all 30 cells. Every seed block favored DQN, and the largest Holm-adjusted sign-test value was 5.722e-05. The least-favorable cell was Double DQN under energy limitation, with a mean difference of -5.92 and a simultaneous 95% upper bound of -5.03 (Table 5; Fig. 1). Nominal DQN-family means spanned 77.66–78.40, compared with 90.83 for MPC-4. This direct inference establishes the DQN-family advantage as the central benchmark finding.

7.2 Target design fine-tunes performance within a consistently strong DQN family

Table 6: Episode total cost across six DQN-family objectives in the five fully coupled regimes. Values are mean $\pm$ SD across 20 independently trained model-seed blocks after within-seed averaging of 20 shared held-out traces. Lower is better; no row is designated as a uniformly preferred method.

Training objective	Nominal	Burst	Link-limited	Energy-limited	Thermal-stress
Standard DQN	78.33 $\pm$ 1.88	97.36 $\pm$ 2.01	102.26 $\pm$ 1.75	119.67 $\pm$ 2.14	97.55 $\pm$ 2.03
Full-action Q auxiliary	78.14 $\pm$ 2.23	97.21 $\pm$ 2.25	102.17 $\pm$ 2.09	119.78 $\pm$ 2.38	97.41 $\pm$ 2.25
Immediate advantage auxiliary	77.84 $\pm$ 1.82	97.22 $\pm$ 2.05	101.95 $\pm$ 1.88	119.17 $\pm$ 2.12	97.40 $\pm$ 2.04
Centered full-action	77.66 $\pm$ 1.84	96.88 $\pm$ 1.94	101.80 $\pm$ 1.80	119.00 $\pm$ 1.65	97.10 $\pm$ 1.93
Double DQN	78.40 $\pm$ 1.98	97.61 $\pm$ 2.15	102.59 $\pm$ 2.08	120.08 $\pm$ 2.41	97.80 $\pm$ 2.12
Double + centered full-action	77.73 $\pm$ 1.99	96.90 $\pm$ 2.02	101.91 $\pm$ 1.88	119.07 $\pm$ 1.77	97.13 $\pm$ 2.01

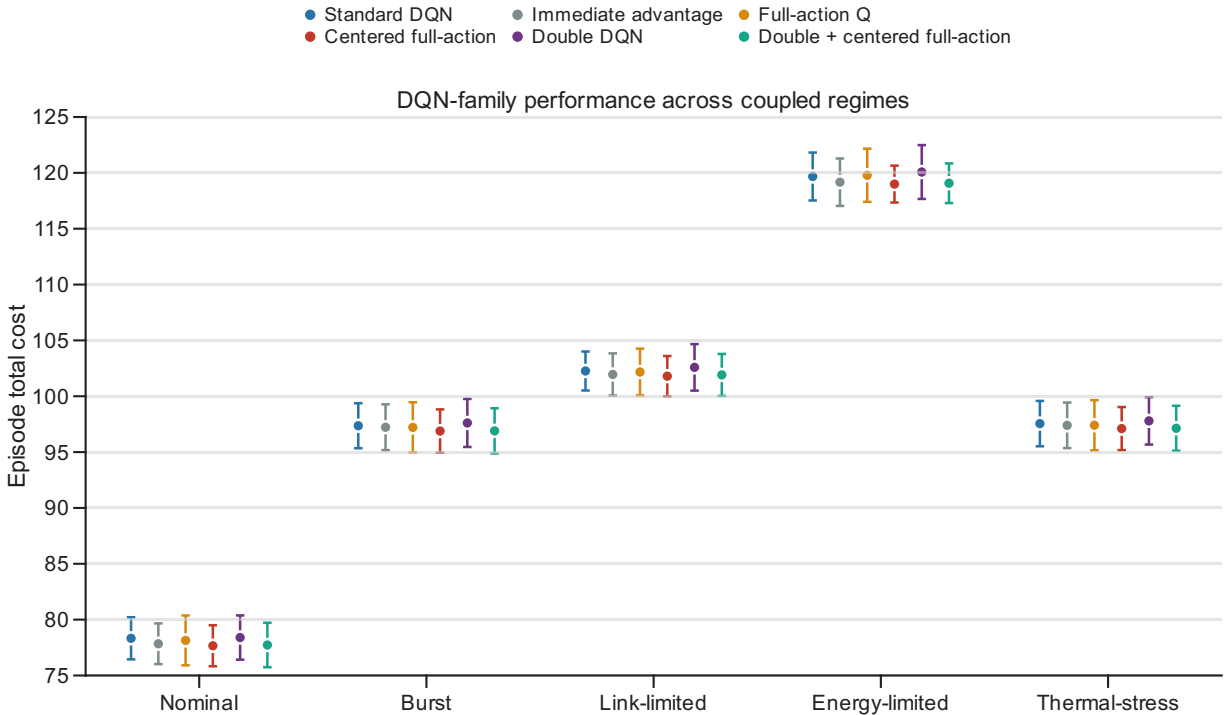


Figure 7: DQN-family performance across fully coupled regimes. Points show model-seed means and error bars show ± 1 SD across 20 independently trained model seeds. The figure emphasizes overlap and scenario-dependent ordering rather than a single winning objective.

Table 7: Nominal fixed-coefficient component and backbone comparisons. All models use 20 pre-specified model-seed blocks, identical real-interaction budgets, and the same held-out traces. Full-action Q and immediate advantage receive the same three modeled action outcomes as the centered full-action variant. The shared $\lambda = 0.3$ was calibrated for the centered objective and not tuned separately for the other auxiliary losses, so these rows do not isolate centering or bootstrapping. Holm adjustment covers the five fully coupled regimes separately for each displayed contrast. “Supported” requires a favorable mean, a paired interval below zero, and Holm $p < 0.05$.

Training objective	Mean cost $\pm$ SD	Matched reference	Paired difference [95% interval]	Holm p	Supported
Standard DQN	78.33 $\pm$ 1.88	—	—	—	—
Full-action Q auxiliary	78.14 $\pm$ 2.23	Standard DQN	-0.19 [-0.56, 0.28]	0.05909	No
Immediate advantage auxiliary	77.84 $\pm$ 1.82	Standard DQN	-0.49 [-0.90, -0.12]	1	No
Centered full-action	77.66 $\pm$ 1.84	Standard DQN	-0.67 [-0.99, -0.39]	0.01288	Yes
Double DQN	78.40 $\pm$ 1.98	—	—	—	—
Double + centered full-action	77.73 $\pm$ 1.99	Double DQN	-0.67 [-1.18, -0.24]	0.03545	Yes

Table 8: Paired inference for centered full-action minus standard DQN across the five fully coupled regimes. Negative differences favor centered full-action supervision. Each comparison uses 20 model-seed blocks after averaging 20 shared held-out traces. Holm adjustment covers the five prespecified regimes, and LOO denotes the leave-one-seed-out mean range.

Regime	Mean difference	95% interval	Negative seeds	Holm p	LOO range
Nominal	-0.67	[-0.99, -0.39]	17/20	0.01288	[-0.71, -0.55]
Burst	-0.48	[-0.73, -0.28]	17/20	0.01288	[-0.51, -0.37]
Link-limited	-0.46	[-0.77, -0.17]	15/20	0.08278	[-0.52, -0.38]
Energy-limited	-0.68	[-1.17, -0.21]	16/20	0.03545	[-0.79, -0.51]
Thermal-stress	-0.45	[-0.75, -0.22]	15/20	0.08278	[-0.48, -0.33]

Table 9: Training information and computation accounting. Wall time is mean $\pm$ SD across 20 models. “Modeled action outcomes” counts analytical action evaluations used by the learning objective. Equal real interactions therefore do not imply equal information or computational budgets.

Training objective	Wall time per model (s)	Real interactions	Updates	Modeled action outcomes	Models
Standard DQN	37.4 $\pm$ 10.4	38,400	9,569	0	20
Full-action Q auxiliary	46.0 $\pm$ 1.4	38,400	9,569	115,200	20
Immediate advantage auxiliary	42.9 $\pm$ 1.1	38,400	9,569	115,200	20
Centered full-action	49.0 $\pm$ 3.1	38,400	9,569	115,200	20
Double DQN	36.0 $\pm$ 1.6	38,400	9,569	0	20
Double + centered full-action	53.2 $\pm$ 3.9	38,400	9,569	115,200	20

Across the five fully coupled regimes, the six objectives occupied a narrow performance band (Table 6; Fig. 7). Nominal means ranged from 77.66 to 78.40. No objective was uniformly best, and the family-to-MPC gap exceeded the differences among target designs.

Centered full-action produced favorable mean differences against standard DQN in all five regimes and met the complete support rule in 3 (Table 8). The remaining component and Double-DQN contrasts changed the ordering by smaller, scenario-dependent amounts (Table 7). Target construction therefore tuned performance without determining the family-level result. All objectives used the same real-interaction budget, while the auxiliary objectives additionally consumed modeled action outcomes (Table 9).

7.3 Deeper exact planning improves cost but retains a gap to the DQN family

Table 10: Nominal receding-horizon planning over 400 unique held-out traces. Each planner has perfect knowledge of the next H tasks and exogenous link states and exactly enumerates 3^H sequences before every action. Runtime was measured on the same workstation and includes only planner action selection after environment construction. DQN inference time was not benchmarked in this experiment, so the table is not a matched deployment-latency comparison.

Planning horizon	Enumerated sequences	Mean cost	Time per decision (ms)	Held-out traces
1	3	99.33	0.13	400
2	9	92.80	0.50	400
4	81	90.83	4.82	400
6	729	89.35	42.50	400

Increasing planning depth reduced nominal mean cost from 99.33 at $H = 1$ to 89.35 at $H = 6$. Over the same range, mean decision time increased from 0.13 to 42.50 ms. Even MPC-6 remained above the nominal DQN-family range of 77.66–78.40. Exact short-horizon planning therefore narrowed, but did not close, the family-level cost gap.

Table 11: Engineering outcome decomposition for standard DQN relative to MPC-4. Standard DQN is shown as a conventional representative of the six-objective family. Delta columns report standard DQN minus MPC-4; negative values denote lower use or cost. Low-energy steps count decisions with remaining energy below 0.18 and are soft-threshold exposures, not certified safety violations.

Regime	Δ immediate	Δ penalty	Δ energy	Δ latency (s)	Δ data (Mbit)
Nominal	0.76	-13.26	-0.104	-1.04	-543.9
Burst	0.42	-11.40	-0.098	-0.92	-489.8
Link-limited	-0.57	-12.91	-0.130	-1.46	-388.1
Energy-limited	1.36	-7.68	-0.102	-0.93	-601.5
Thermal-stress	0.32	-11.97	-0.100	-0.97	-482.5

Regime	Low-energy steps DQN/MPC-4	Min. energy DQN/MPC-4	Max. heat DQN/MPC-4
Nominal	23.6/28.5	0.00/0.00	0.192/0.190
Burst	28.8/32.3	0.00/0.00	0.192/0.190
Link-limited	30.2/35.0	0.00/0.00	0.193/0.191
Energy-limited	38.8/41.8	0.00/0.00	0.193/0.190
Thermal-stress	28.8/32.4	0.00/0.00	0.364/0.362

Table 11 decomposes the family-planner gap using standard DQN as a conventional family member. Relative to MPC-4, standard DQN reduced delayed penalties in every fully coupled regime while changing immediate cost by only -0.57 to 1.36 . It also used less modeled energy, latency, and transmitted data. Both methods reached zero in the clipped energy state, while standard DQN produced slightly higher maximum heat.

7.4 Representative DQN training diagnostics show empirical stabilization

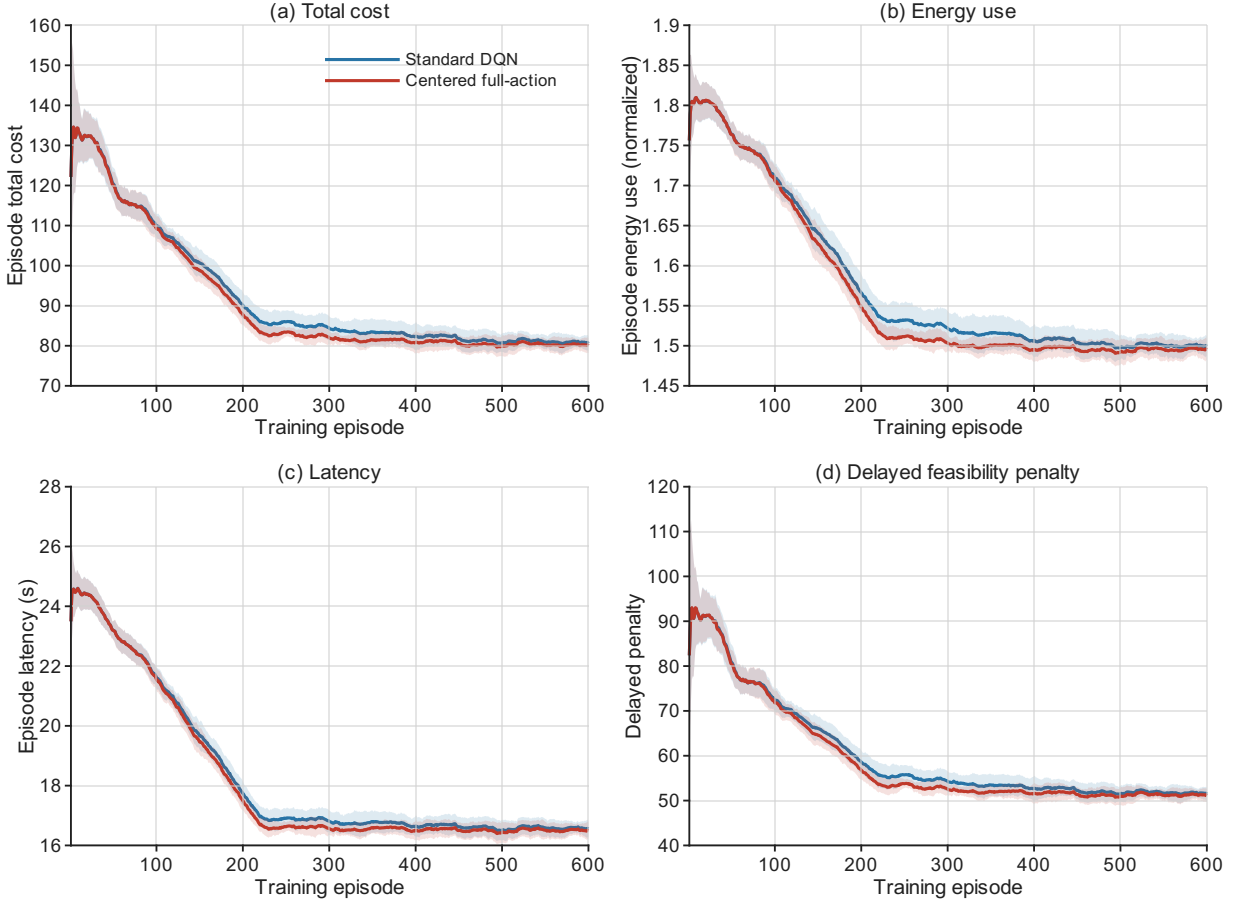


Figure 8: Representative training diagnostics for standard DQN and the centered full-action DQN variant. Lines are means across 20 model seeds after a causal 25-episode moving average; shaded regions are ± 1 SD. Curves describe the training distribution and are not used as independent test observations or evidence for the whole DQN family.

All four logged quantities declined during approximately the first 220 episodes and then fluctuated around stable ranges (Fig. 8). The late-episode trajectories provide evidence of empirical optimization stability for the representative DQN members.

7.5 Within-family sensitivity under parameter changes

Table 12: Prespecified parameter sensitivity for centered full-action minus standard DQN. Negative differences favor the centered full-action variant; $n = 20$ independent paired model seeds in each profile. Holm adjustment covers the eight displayed profiles.

Parameter profile	Mean difference	95% interval	Improvement	Holm p
Reference	-0.67	[-0.99, -0.39]	0.85%	0.01804
Compute demand -20%	-0.49	[-0.78, -0.22]	0.79%	0.01804
Compute demand $+20\%$	-0.89	[-1.35, -0.53]	0.98%	0.00322
Input data -20%	-1.03	[-1.55, -0.62]	1.66%	0.01804
Input data $+20\%$	-0.44	[-0.69, -0.23]	0.50%	0.01804
Energy use $+20\%$	-0.60	[-0.98, -0.28]	0.57%	0.01804
Heat load $+20\%$	-0.67	[-1.01, -0.40]	0.86%	0.01804
Link capacity -20%	-0.64	[-0.96, -0.35]	0.72%	0.01804

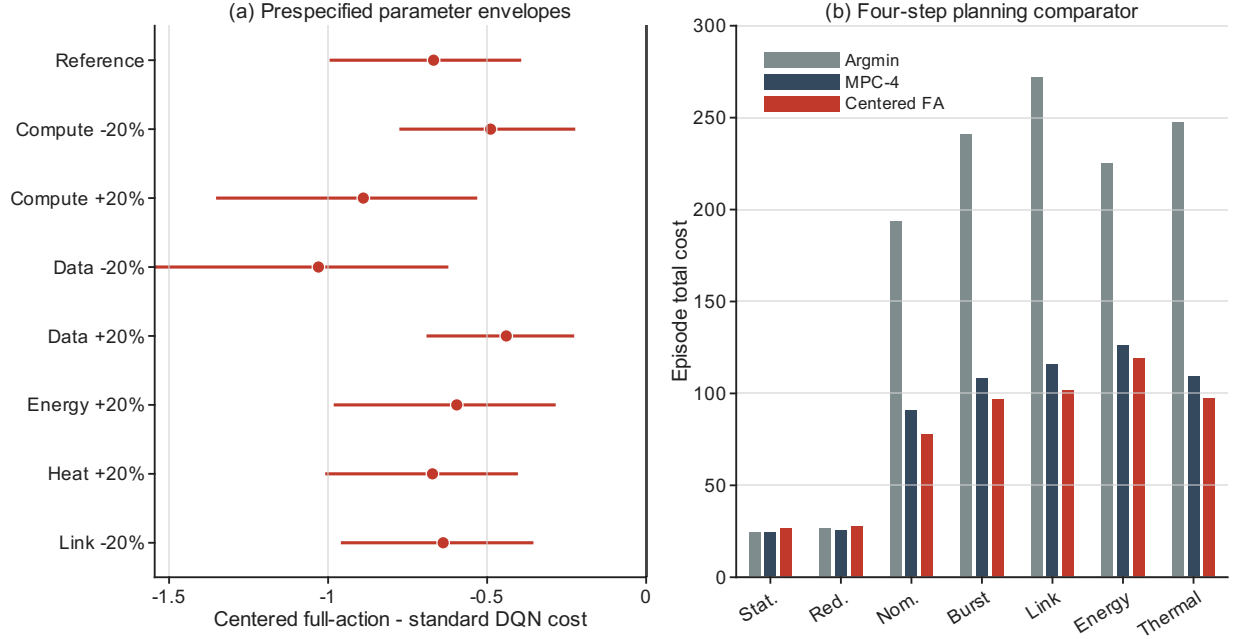


Figure 9: Parameter sensitivity and planning boundary. (a) Mean paired centered full-action–standard DQN differences with 95% paired bootstrap intervals. (b) Immediate, four-step rolling, and learned scheduling across coupling and stress regimes.

The paired centered-full-action–standard-DQN interval remained below zero in all eight profiles, with relative differences from 0.50% to 1.66%. The largest difference occurred when raw data volume decreased by 20%, whereas the smallest occurred when it increased by 20%. The within-family direction therefore remained stable across the prespecified parameter envelope.

7.6 Preview-scale perturbations bound one model-assisted variant

Table 13: Effect of training-time preview misspecification. Differences are relative to the same standard-DQN models evaluated in the factual simulator. Scaling applies only to the centered full-action variant’s training-time preview cost and action-induced resource changes.

Training preview	Regime	Mean difference	95% interval	Negative seeds	Holm p	Supported
Simulator-exact preview	Nominal	-0.67	[-0.99, -0.39]	17/20	0.01288	Yes
	Burst	-0.48	[-0.73, -0.28]	17/20	0.01288	Yes
	Link-limited	-0.46	[-0.77, -0.17]	15/20	0.08278	No
	Energy-limited	-0.68	[-1.17, -0.21]	16/20	0.03545	Yes
	Thermal-stress	-0.45	[-0.75, -0.22]	15/20	0.08278	No
Consequences scaled to 0.85	Nominal	-0.45	[-0.85, -0.12]	18/20	0.002012	Yes
	Burst	-0.41	[-0.72, -0.18]	17/20	0.01031	Yes
	Link-limited	-0.37	[-0.62, -0.13]	13/20	0.5264	No
	Energy-limited	-0.27	[-0.71, 0.14]	13/20	0.5264	No
	Thermal-stress	-0.37	[-0.70, -0.10]	16/20	0.03545	Yes
Consequences scaled to 1.15	Nominal	-0.77	[-1.11, -0.49]	20/20	9.537e-06	Yes
	Burst	-0.48	[-0.74, -0.29]	19/20	0.0001602	Yes
	Link-limited	-0.33	[-0.62, -0.06]	13/20	0.2632	No
	Energy-limited	-0.92	[-1.45, -0.46]	17/20	0.00773	Yes
	Thermal-stress	-0.42	[-0.71, -0.18]	15/20	0.08278	No

Uniform 15% under- and over-scaling preserved favorable centered-full-action differences (Table 13). The two conditions met the complete rule in 3 and 3 regimes, respectively. Their largest Holm-adjusted sign-test values were 0.5264 and 0.2632. Scalar preview perturbations therefore did not reverse the observed within-family direction.

8 Discussion

The principal contribution is a family-level result rather than the success of one selected objective. Six DQN objectives retained a common advantage over MPC-4 despite changes in supervision, centering, bootstrapping, and Double-DQN evaluation. Their narrow internal spread further shows that the shared sequential value-learning formulation dominated the smaller effects of target design.

The contextual bandit comparison and planning-depth analysis support a temporal interpretation of this result. Immediate-reward learning did not reproduce the DQN-family performance, while deeper exact enumeration progressively narrowed the cost gap. Together, these observations are consistent with learned values capturing consequences that persist beyond the current allocation.

The benchmark provides a second contribution by making that comparison traceable and reusable. Shared task primitives generate heat, energy, latency, and transmitted volume, while transition audits connect the analytical previews to executed outcomes. Scripts then link checkpoints and seed-level estimates to tables, figures, and validation hashes. This evidence chain turns the scheduling problem into a reproducible platform for comparing sequential decision objectives.

8.1 Future work

The first priority is operational validation. The current simulator uses dimensionless fixed weights and clipped soft resource states, so future versions should enforce feasibility through action masking, constrained reinforcement learning, and terminal resource conditions. Mission-derived utilities,

telemetry, and hardware-in-the-loop traces can then test the ordering under physical resource dynamics. Multi-satellite, multi-ground-station, and variable-contact topologies will further establish scalability.

The second priority is causal separation of the mechanisms that produce the family-level pattern. Future experiments should tune auxiliary weights by objective, normalize gradient contributions, and factorially separate centering, Bellman bootstrapping, and discounting. Transition carryover and delayed-penalty scaling should also vary independently rather than share the coupling coefficient c . Regime-specific retraining will complement the current $c = 1$ transfer tests. Structural, state-dependent, and action-specific preview errors, together with family-wide action-mixture analyses, can identify which components sustain the result.

The third priority is a broader and computationally matched comparator frontier. Scalable tree search, dynamic programming, mixed-integer optimization, and satellite-specific schedulers should be evaluated under matched information and computation budgets. DQN inference latency and energy should be measured on the same deployment path as planning. These studies will determine how the benchmark advantage transfers to operational scheduling.

9 Conclusion

We establish a statistically supported DQN-family advantage in a reproducible resource-coupled satellite-ground scheduling benchmark. All 30 paired objective–regime contrasts favored DQN over MPC-4, with a family-wide simultaneous 95% upper bound of -5.03. Differences among DQN objectives were smaller and scenario dependent, making family-level consistency the principal contribution. The benchmark now provides a rigorous foundation for the mission-calibrated, hard-constrained, and computationally matched studies defined in Future work.

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Data and Code Availability

The local versioned research package contains the simulator, checkpoints, raw evaluations, seed-level summaries, direct DQN–MPC-4 inference, Python and MATLAB figure source, and LaTeX-table generators under `md_cbad_dqn/` and `paper/figures/`. It includes the primary, preview-error, component-control, Double-DQN, joint-bootstrap, and maximum-statistic results. Machine-readable validation and metadata files record expected rows, model seeds, training budgets, randomization blocks, package versions, source files, and artifact hashes. The commands in the reproducibility subsection regenerate the result chain. A permanent public archive has not yet been supplied for review. **AUTHOR ARCHIVE IDENTIFIER TO BE INSERTED BEFORE SUBMISSION.**

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